GaugeHD Synthetic — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — GaugeHD Synthetic (MKM-GHD-001)

Tests run on **01 April 2026 at 10:24:44**.

Table 1: Test Results Summary — GaugeHD Synthetic (MKM-GHD-001)

Total Tests	62
Passed	62
Failed	0
Skipped	0
Duration	8.19s

Table 2: Detailed Test Results — GaugeHD Synthetic

Test Description	Acceptance Criteria	Result	Time
_snap_to_river skips zero-length segments without error.	`isinstance(lat, float); isinstance(lon, float)`	Pass	0.000s
_create_synthetic_gauge returns None when flanking gauge not in lookup.	`result is None`	Pass	0.004s
Lines 83-110: generate_all_gauge_histories processes all gauges.	`isinstance(result, list); len(result) == 1`	Pass	0.225s
Lines 105-107: error on one gauge -¿ logged, continues.	`isinstance(result, list)`	Pass	0.379s
All levels positive	`all(obs["level_meters"] > 0 for obs in result)`	Pass	0.006s
Correct length approximately	`728 <= len(result) <= 732`	Pass	0.003s
Dates are sequential	`(d_curr - d_prev).days == 1`	Pass	0.003s
Different seeds different output	`levels1 != levels2`	Pass	0.012s
Each obs has date and level	`"date" in obs; "level_meters" in obs` (+1 more)	Pass	0.001s
Gauge data without FloodGauge wrapper uses defaults.	`isinstance(result, list)`	Pass	0.001s
When historical high date is within range, that day gets historical_high level.	`match[0]["level_meters"] == 7.5`	Pass	0.001s
Invalid date format doesn't crash.	`isinstance(result, list)`	Pass	0.001s
Historical high date before start_date is ignored.	`isinstance(result, list)`	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
Lines 176-179: Historical high injected via fallback when date not matched in loop.	`len(match) == 1; match[0]["level_meters"] == 8.0`	Pass	0.001s
Gauge data with no FloodStages uses sensible defaults.	`isinstance(result, list); len(result) > 0`	Pass	0.001s
Lines 97-100: Partially tidal gauge uses dampened tidal signal.	`len(result) > 700; all(lev > 0 for lev in levels)`	Pass	0.003s
Returns list	`isinstance(result, list)`	Pass	0.001s
Seed reproducible	`r1 == r2`	Pass	0.003s
Lines 92-95: Tidal gauge uses wider daily_std, tidal_amplitude ¿ 0.	`len(result) > 700; max(levels) > min(levels)`	Pass	0.003s
Lines 147-149: Tidal component with spring-neap cycle.	`l1 != l2`	Pass	0.003s
Lines 138-141: historical high injected after main loop.	`len(match) == 1; match[0]["level_meters"] == 9.99`	Pass	0.003s
When historical_high_dt is None (out of range), post-loop block skipped.	`all(o["level_meters"] != 8.0 for o in result)`	Pass	0.002s
When historical_high_date is None, both branches skipped.	`isinstance(result, list); len(result) > 0`	Pass	0.001s
generate() returns count=0 when GAUGE_POINTS has ¡ 2 entries.	`result["count"] == 0`	Pass	0.001s
generate() returns count=0 when GAUGE_POINTS is None.	`result["count"] == 0`	Pass	0.001s
Lines 74-75: missing file raises FileNotFoundError.	—	Pass	0.000s
Lines 72-80: load_gauge_portfolio reads gauge.json.	`isinstance(result, list); len(result) == 1`	Pass	0.002s
_load_river_polyline returns None when cache file doesn't exist.	`result is None or isinstance(result, list)`	Pass	0.001s
_snap_to_river returns original coords when river is None.	`lat == 51.46; lon == -0.30`	Pass	0.000s
Lines 143-144: bad CSV -¿ error logged, continues.	`isinstance(result, list)`	Pass	0.002s
Lines 113-147: process_nrfa_directory scans CSV files.	`isinstance(result, list); len(result) == 1`	Pass	0.002s
Line 134: output_dir=None uses config.get_gaugehd_dir().	`isinstance(result, list)`	Pass	0.003s
Lines 131-147: no matching files -¿ empty list.	`result == []`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Property projecting to seg_idx=0, t˜0 should not create a gauge.	`result["count"] == 0`	Pass	0.003s
Propertyhc has synth in nearest gauges	`found_synth`	Pass	0.008s
Synthetic gauge in propertyhc should have basis_bps computed.	`"basis_bps" in ng`	Pass	0.006s
Every property has one synthetic gauge	`not failures`	Pass	1.909s
Every property has three gauges total	`len(ngs) == 3`	Pass	0.154s
Every property has two real gauges	`not failures`	Pass	1.676s
Properties should have flood_events sourced from gauge storm responses.	`properties_with_events > 0`	Pass	1.674s
At least some properties should have flooded=True events.	`flooded_count > 0`	Pass	1.667s
Appends to gauge json	`len(data["flood_gauges"]) > 3`; `len(synth) > 0`	Pass	0.011s
Creates synthetic gauges	`result["count"] > 0`; `len(result["gauge_ids"]) == result["count"]`	Pass	0.015s
Two properties near the same river point should share a synthetic gauge.	`result["count"] == 2`	Pass	0.015s
Property far beyond polyline end should not create a synthetic gauge.	`result["count"] == 0`	Pass	0.005s
Interpolated elevation	`3.0 <= elev <= 5.0`	Pass	0.010s
Interpolated flood stages	`3.5 <= stages["FloodAlert"] <= 4.5`; `4.5 <= stages["FloodWarning"] <= 5.5` (+1 more)	Pass	0.011s
No gauge json returns zero	`result["count"] == 0`	Pass	0.000s
No property json returns zero	`result["count"] == 0`	Pass	0.003s
Synthetic gauge has cdm structure	`"GaugeID" in fg["Header"]`; `"CatchmentID" in fg["Header"]` (+8 more)	Pass	0.009s
Synthetic gauge lat/lon should be on the polyline (very close to it).	`dist_m < 50.0`; `dist_m < 1.0`	Pass	0.015s
A pre-existing SYNTH- gauge in gauge.json is skipped in polyline matching.	`result["count"] >= 0`	Pass	0.010s
Gauge json has synth gauges	`len(synth) > 0`	Pass	0.002s

Test Description	Acceptance Criteria	Result	Time
Synth gauge has required cdm fields	`fg["Header"]["GaugeID"].startswith("SYNTH")`; `fg["Location"]["GaugeLatitude"] != 0` (+7 more)	Pass	0.002s
Gaugehc has synth curves	`len(synth) > 0`	Pass	0.013s
Synth has term structure	`len(ts) > 0`	Pass	0.007s
Synth hazard rates positive	`hc.get("annual_hazard_rate_alert", 0) > 0`	Pass	0.007s
Synth gaugets files exist	`len(synth_files) > 0`	Pass	0.001s
At least some storms should exceed alert at synthetic gauges.	`len(exceeded) > 0`	Pass	0.135s
Synth gaugets have storm responses	`len(resps) > 0`	Pass	0.150s
Property with lat=0 is silently skipped.	`result["count"] == 0`	Pass	0.002s
Property with lon=0 is silently skipped.	`result["count"] == 0`	Pass	0.003s

1.1 Test Document History

Date	Version	Description	Author
27-Mar-2026	1.0	Initial test suite (80 tests)	D. Kelly
01-Apr-2026	1.1	18 test(s) removed (total: 62)	D. Kelly